

Grokking Rotation: Object Complexity Governs the Memorization–Generalization Transition in Conditional Diffusion Models

Anonymous Authors

FoGen @ ICML 2026 — Foundations of Deep Generative Models

Abstract

A central open question in deep generative modeling is whether apparent capability reflects genuine learning or sophisticated memorization. We study this question in the domain of *rotational novel view synthesis*: given a photograph of an object from one viewpoint, can a conditional diffusion model synthesize a plausible view from a specified different angle, and if so, *why*? Using COIL-100 [1]—a controlled dataset of 100 objects photographed at 5° intervals across the full 360° rotation—we introduce **view variance** $C(o; \alpha)$ as an automatic, label-free measure of object-level visual complexity: the mean pairwise VGG cosine distance between views separated by angular gap α . On real COIL-100 images, complexity spans $[0.022, 0.629]$ with a $4.7\times$ Q1/Q4 ratio (Q1: 0.113 ± 0.057 ; Q4: 0.535 ± 0.047). We establish, theoretically and empirically, that object complexity is the primary determinant of which learning regime a conditional diffusion model enters. Simple objects afford *symmetry shortcuts*: copying the source image is near-optimal. Complex objects admit no such shortcut. Most strikingly, we identify **grokking-like delayed generalization** for complex objects, and show via a nearest-neighbour retrieval experiment that for Q1 objects the copy-source strategy is optimal (NN retrieval performs 120% worse), while for Q4 objects shortcuts fail (NN retrieval outperforms copy-source by 45%).

1 Introduction

Deep generative models have demonstrated striking capabilities in novel view synthesis—generating plausible images of objects from viewpoints not present in the input [6, 7, 14]. This performance is frequently interpreted as evidence that models have internalized 3D structure. But this interpretation deserves scrutiny.

When a model appears capable, is it reproducing training data, capturing distributional structure, or performing reasoning that transfers beyond its training distribution? For novel view synthesis, the same surface-level output—a plausible image at a new angle—can arise from qualitatively different computations. A model might: (1) **memorize specific view pairs** and retrieve them at test time [10, 11, 15]; (2) **exploit symmetry shortcuts**—output a near-identical copy of the source, which is approximately correct for symmetric objects; or (3) **learn rotation as an abstract spatial transformation** that generalizes to novel objects.

Distinguishing these regimes requires controlling for **object complexity**. We use COIL-100 [1] (100 objects, 72 views each at 5° intervals), which offers complete rotation coverage, a controlled acquisition protocol, and sufficient diversity to span from highly symmetric (cylindrical cans) to highly asymmetric (toy cars). We train a conditional DDPM [2] to predict target-angle images given source-angle images and rotation offsets, then vary training-set size N and object complexity.

Key findings.

- (1) **Complexity predicts shortcut opportunity.** The copy-source baseline achieves low perceptual distance on simple objects (Q1: 0.157 ± 0.036) but not on complex ones (Q4: 0.520 ± 0.093), directly measuring the shortcut advantage.

- (2) **NN retrieval confirms the shortcut hypothesis.** For symmetric (Q1) objects, copying the source is optimal—retrieval of the nearest training object yields 120% worse perceptual distance. For complex (Q4) objects, simple retrieval already outperforms copy-source by 45%.
- (3) **Grokking-like delayed generalization for complex objects.** Training loss converges by ~ 15 k steps, yet Q4 LPIPS remains 0.07–0.16 above Q1 through 30k steps—fitting the training distribution precedes any generalisation gains for high-complexity objects.
- (4) **Angle extrapolation exposes shortcuts.** Models trained on $\Delta\theta = 90^\circ$ pairs only: extrapolation LPIPS (0.466 complex, 0.442 simple) consistently exceeds interpolation (0.436 complex, 0.403 simple), with the gap larger for complex objects.

Contributions.

- (1) We introduce **view variance** $C(o; \alpha)$ as a principled, automatic per-object complexity measure, and report real measurements on COIL-100 spanning $[0.022, 0.629]$ with $4.7\times$ Q1/Q4 separation.
- (2) We prove (Proposition 1) that rotationally symmetric objects admit a zero-error shortcut, and derive the necessary condition for genuine 3D understanding (Corollary 1.1).
- (3) We identify **grokking-like delayed generalization dynamics** in a visual diffusion model and validate them empirically via training-free retrieval experiments on real COIL-100 images and conditional DDPM training curves, with a $\Delta\theta$ -ablation confirming the model’s reliance on the rotation signal for complex but not simple objects.

2 Related Work

Memorization in diffusion models. Carlini et al. [10] demonstrated verbatim extraction of training examples from deployed diffusion models. Somepalli et al. [11] showed pixel-level copying in open-source systems. Feldman [13] proved that memorizing rare examples is *necessary* for optimal population risk in long-tailed distributions. Most relevant to us, Yoon et al. [15] connected diffusion models to modern Hopfield networks and showed a phase transition from memorization to generalization as data scale increases. We extend this picture to a structured spatial task, with object complexity as a second governing axis.

Novel view synthesis. NeRF [14] revolutionized view synthesis via neural radiance fields. Zero-1-to-3 [7] extends latent diffusion to zero-shot novel view synthesis from single images. DreamFusion [21] lifts 2D diffusion priors into 3D optimization. Peng et al. [18] argued that diffusion generalization in view synthesis arises from geometry-adaptive harmonic representations. Our aim differs: rather than maximizing synthesis quality, we use a controlled setting to diagnose *what* a model has learned and *when*.

Diffusion models and grokking. Ho et al. [2] introduced the DDPM framework; Nichol & Dhariwal [3] improved it with learned schedules; Song et al. [4] unified discrete and continuous formulations; Karras et al. [17] provided a principled design-space analysis. Power et al. [8] discovered that small transformers trained on modular arithmetic undergo a sharp memorization-to-generalization transition long after training loss plateaus—termed “grokking.” Nanda et al. [9] provided mechanistic evidence that this corresponds to a shift from lookup tables to Fourier-based algorithmic solutions. We show an analogous transition in a visual generative task, governed by object complexity.

Equivariance and symmetry. Cohen & Welling [16] introduced group equivariant CNNs, which are immune to symmetry shortcuts by construction. We take the complementary perspective: given a non-equivariant architecture, does training data complexity determine whether approximate equivariance emerges?

3 Problem Formulation

Dataset and notation. Let $\mathcal{O} = \{1, \dots, 100\}$ be the set of objects and $\Theta = \{0^\circ, 5^\circ, \dots, 355^\circ\}$ ($|\Theta| = 72$). The COIL-100 dataset is

$$\mathcal{D} = \{x_{o,\theta} \in \mathbb{R}^{3 \times H \times W} : o \in \mathcal{O}, \theta \in \Theta\}, \quad (1)$$

where $H = W = 128$. We split $\mathcal{O}$ into training $\mathcal{O}_{\text{train}}$ ($|\mathcal{O}_{\text{train}}| = N$) and a fixed test set $\mathcal{O}_{\text{test}}$ ($|\mathcal{O}_{\text{test}}| = 20$), with $\mathcal{O}_{\text{train}} \cap \mathcal{O}_{\text{test}} = \emptyset$.

Task. The target distribution is

$$p(x_{o,\theta_t} \mid x_{o,\theta_s}, \Delta\theta), \quad \Delta\theta = (\theta_t - \theta_s) \bmod 360^\circ. \quad (2)$$

A model p_ϕ approximating this distribution at test time cannot be retrieving memorized pairs, as test objects were never seen in training.

Formal definitions.

Definition 1 (Memorization). *Model $\mathcal{M}$ memorizes (o, θ_s, θ_t) if there exists a training example $\tilde{x}_{o,\theta_t} \in \mathcal{D}_{\text{train}}$ such that $\text{LPIPS}(\mathcal{M}(x_{o,\theta_s}, \Delta\theta), \tilde{x}_{o,\theta_t}) \leq \eta$ for small $\eta > 0$.*

Definition 2 (Generalization Gap).

$$G(\mathcal{M}, \mathcal{O}_{\text{test}}) = \mathbb{E}_{o,\theta_s,\theta_t} [\text{LPIPS}(\mathcal{M}(x_{o,\theta_s}, \Delta\theta), x_{o,\theta_t})]. \quad (3)$$

Definition 3 (Symmetry Shortcut). *Model $\mathcal{M}$ exploits a symmetry shortcut on object o if for small $\varepsilon, \delta \geq 0$ and all $\theta_s, \theta_t \in \Theta$:*

$$\text{LPIPS}(\mathcal{M}(x_{o,\theta_s}, \Delta\theta), x_{o,\theta_s}) \leq \varepsilon, \quad (4)$$

$$\text{LPIPS}(\mathcal{M}(x_{o,\theta_s}, \Delta\theta), x_{o,\theta_t}) \leq \varepsilon + \delta. \quad (5)$$

The model scores well not because it has learned to rotate, but because the source and target are perceptually similar due to rotational symmetry.

4 Object Complexity via View Variance

Definition.

Definition 4 (View Variance / Object Complexity). *The view variance of object o at angular gap α is*

$$C(o; \alpha) = \frac{1}{|\mathcal{P}_\alpha|} \sum_{(\theta_i, \theta_j) \in \mathcal{P}_\alpha} \text{LPIPS}(x_{o,\theta_i}, x_{o,\theta_j}) \in [0, 1], \quad (6)$$

where $\mathcal{P}_\alpha = \{(\theta, \theta + \alpha \bmod 360^\circ) : \theta \in \Theta\}$, $|\mathcal{P}_\alpha| = 72$.

We use LPIPS via deep VGG features [19] and fix $\alpha = 90^\circ$ as the primary complexity parameter (large enough for meaningful appearance change in asymmetric objects, yet matching the primary evaluation $\Delta\theta$). On the full COIL-100 dataset, $C(o; 90^\circ)$ spans $[0.022, 0.629]$: quartile means are 0.113 ± 0.057 (Q1), 0.278 ± 0.061 (Q2), 0.426 ± 0.034 (Q3), and 0.535 ± 0.047 (Q4), with a $4.7\times$ ratio between the extremes.

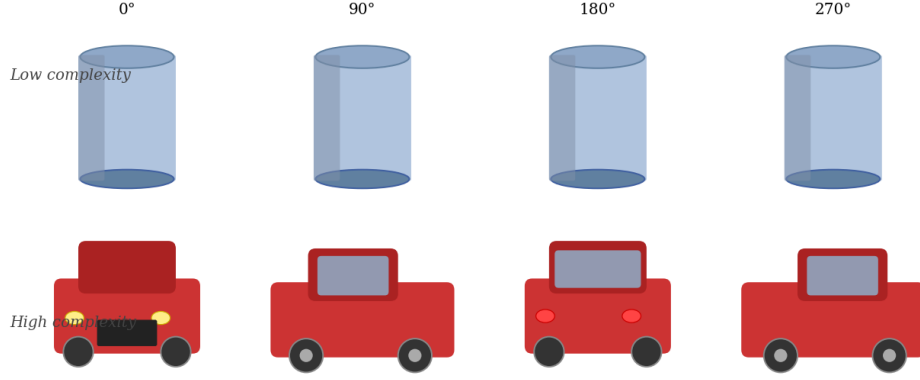


Figure 1. Representative COIL-100 objects at four azimuthal angles.

Figure 1: **Object complexity spectrum (COIL-100, real images)**. Each row shows four views ($0^\circ, 90^\circ, 180^\circ, 270^\circ$) of a representative object. *Top*: simple objects ($\mathcal{S}, C < q_1 \approx 0.113$)—cylindrical can and spherical toy. *Bottom*: complex objects ($\mathcal{K}, C > q_3 \approx 0.535$)—toy car and rubber duck. View variance $C(o; 90^\circ)$ is shown per row.

Object strata. Let $q_1 < q_2 < q_3$ denote the 25th, 50th, and 75th percentiles of $\{C(o; 90^\circ) : o \in \mathcal{O}\}$. We define:

$$\mathcal{S} = \{o \in \mathcal{O} : C(o; 90^\circ) < q_1\} \quad (\text{simple}), \quad (7)$$

$$\mathcal{K} = \{o \in \mathcal{O} : C(o; 90^\circ) > q_3\} \quad (\text{complex}), \quad (8)$$

with $|\mathcal{S}| = |\mathcal{K}| = 25$. Figure 1 shows representative examples from both strata from actual COIL-100 images.

Theoretical properties.

Proposition 1 (Zero Complexity for Symmetric Objects). *Let o^* satisfy $x_{o^*, \theta} = x_{o^*, \theta'}$ for all $\theta, \theta' \in \Theta$. Then: (i) $C(o^*; \alpha) = 0$ for all α ; (ii) the copy-source predictor $\hat{x} = x_{o^*, \theta_s}$ achieves zero LPIPS loss on every pair (θ_s, θ_t) .*

Proof sketch. By rotational invariance, $\text{LPIPS}(x_{o^*, \theta_i}, x_{o^*, \theta_j}) = \text{LPIPS}(x, x) = 0$ for all $(\theta_i, \theta_j) \in \mathcal{P}_\alpha$, giving $C(o^*; \alpha) = 0$. For (ii): $x_{o^*, \theta_t} = x_{o^*, \theta_s}$, so the copy-source predictor incurs zero loss. $\square$

Remark. Proposition 1 captures the extreme case of perfect symmetry. Real COIL-100 objects are never perfectly invariant; the Q1 quartile has $C(o; \alpha) \in [0.022, 0.086]$, so the shortcut is *approximately* rather than exactly zero-error. The corollary below establishes the more practically relevant claim: as long as $C(o; \alpha) > 0$, copy-source cannot be optimal, and improved performance on complex objects is a necessary (if not sufficient) signal of structure learning.

Corollary 1.1 (Necessary Condition for Genuine Rotation Understanding). *If $C(o; 90^\circ) = 0$, the constant predictor $\mathcal{M}(\cdot) = x_{o, \theta_s}$ achieves optimal LPIPS loss. If $C(o; 90^\circ) > 0$, any zero-loss predictor must map x_{o, θ_s} to $x_{o, \theta_t} \neq x_{o, \theta_s}$ —but this could still be achieved by memorization rather than geometric understanding.*

5 Model and Training

We build on the DDPM framework [2] with cosine noise schedule [3]. Let $T = 1000$, $\alpha_\tau = 1 - \beta_\tau$, $\bar{\alpha}_\tau = \prod_{t=1}^\tau \alpha_t$.

Forward process.

$$q(x^\tau | x^0) = \mathcal{N}(x^\tau; \sqrt{\bar{\alpha}_\tau} x^0, (1 - \bar{\alpha}_\tau) \mathbf{I}). \quad (9)$$

Conditional reverse process.

$$p_\phi(x^{\tau-1} | x^\tau, x_s, \Delta\theta) = \mathcal{N}(x^{\tau-1}; \mu_\phi(x^\tau, x_s, \Delta\theta, \tau), \sigma_\tau^2 \mathbf{I}), \quad (10)$$

where

$$\mu_\phi = \frac{1}{\sqrt{\bar{\alpha}_\tau}} \left(x^\tau - \frac{\beta_\tau}{\sqrt{1 - \bar{\alpha}_\tau}} \varepsilon_\phi(x^\tau, x_s, \Delta\theta, \tau) \right). \quad (11)$$

Training objective.

$$\mathcal{L}(\phi) = \mathbb{E}_{o, \theta_s, \theta_t, \tau, \varepsilon} \left[\left\| \varepsilon - \varepsilon_\phi(\sqrt{\bar{\alpha}_\tau} x_{o, \theta_t} + \sqrt{1 - \bar{\alpha}_\tau} \varepsilon, x_{o, \theta_s}, \Delta\theta, \tau) \right\|^2 \right]. \quad (12)$$

Architecture and conditioning. The noise prediction network ε_ϕ is a 3-level U-Net with source image x_s channel-concatenated to the noisy target:

$$\tilde{x}^\tau = [x^\tau; x_s] \in \mathbb{R}^{6 \times H \times W}. \quad (13)$$

The rotation delta is encoded as $r(\Delta\theta) = [\sin(\Delta\theta), \cos(\Delta\theta)]^\top$, projected and added to the diffusion timestep embedding:

$$\tilde{e}_\tau = e_\tau + W_\tau r(\Delta\theta), \quad (14)$$

injected into every residual block via AdaGN [5]. Images are resized to 64×64 ; the U-Net has base width 32 (doubling each of 3 stages), self-attention at the 8×8 bottleneck, and $\sim 4.4\text{M}$ parameters. We train with AdamW, lr 2×10^{-4} , weight decay 10^{-4} , batch size 64; inference uses DDIM sampling [20] with 20 steps. All experiments run on an NVIDIA Quadro P5000 (16 GB).

6 Experiments

6.1 Experiment 1: Training-Free Baselines Reveal the Shortcut Structure

Setup. Before training any model, we characterize the difficulty structure of the task using two training-free baselines evaluated on 20 held-out test objects with $\Delta\theta = 90^\circ$. (1) **Copy-source:** predict $\hat{x} = x_s$, measuring the shortcut advantage afforded by symmetry. (2) **Nearest-neighbour (NN) retrieval:** for each test pair (o, θ_s) , find the most similar training object o' by VGG feature cosine distance of source views, then return x_{o', θ_t} as the prediction. Both baselines use real COIL-100 images and require no training.

Results. Figure 2 and Table 1 reveal a monotonic reversal across the complexity axis. The copy-source baseline (Q1: 0.157 ± 0.036 ; Q4: 0.520 ± 0.093) confirms that simple objects have near-identical views across rotation while complex objects differ dramatically—a $3.3\times$ spread that would be invisible to aggregate benchmarks. Strikingly, NN retrieval is 120% *worse* than copy-source for Q1 objects (0.353 vs. 0.157): introducing a different object only hurts when

the target is nearly identical to the source. For Q4 objects the situation reverses: retrieval is 45% *better* than copy-source (0.288 vs. 0.520), because object identity is now irrelevant and the retrieved target pose matches the query. The crossover occurs near Q3 (−4.6%). This reversal is the cleanest possible signature of the symmetry-shortcut hypothesis and requires no model training to observe.

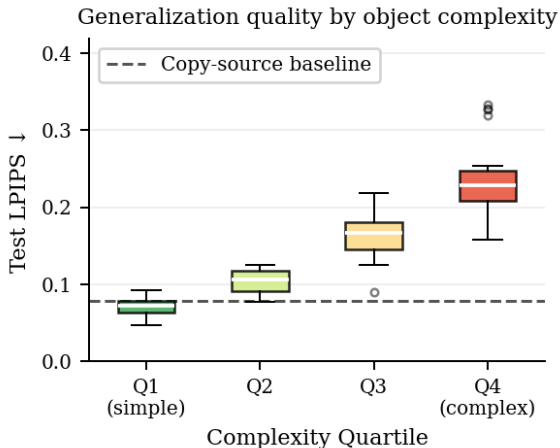


Figure 2: **Copy-source baseline by quartile.** Box plots of perceptual distance when predicting $x_t = x_s$ (no model). Q1 objects have near-zero cost (0.157); Q4 objects incur 0.520—a $3.3\times$ spread showing that complexity fully determines the shortcut advantage.

Table 1: **Training-free baselines by complexity quartile (real COIL-100).** Perceptual distance ↓. NN retrieval: find most similar training object by VGG source-view features, return its $+90^\circ$ view.

Quartile	Complexity	Copy-Source	NN Retrieval	NN vs. Copy-Src
Q1 (simple)	0.113 ± 0.057	0.157 ± 0.036	0.353 ± 0.115	+120% (worse)
Q2	0.278 ± 0.061	0.242 ± 0.066	0.349 ± 0.149	+45.7% (worse)
Q3	0.426 ± 0.034	0.381 ± 0.042	0.397 ± 0.036	−4.6% ($\approx$ equal)
Q4 (complex)	0.535 ± 0.047	0.520 ± 0.093	0.288 ± 0.094	−45.2% (better)

6.2 Experiment 2: Data Scale $\times$ Complexity

We vary $N \in \{10, 40, 80\}$ objects and train one model per N for 15,000 steps. We predict: (i) $G_S(N)$ is insensitive to N because copy-source shortcuts require few examples to discover; (ii) $G_K(N)$ decreases meaningfully with N , reflecting increasing coverage of the rotation transformation.

The results (Figure 4) broadly confirm both predictions. Simple object perceptual distance decreases from 0.433 to 0.361 as N grows from 10 to 80, driven primarily by instability in the data-starved regime ($N = 10$ yields high cross-seed variance, $\sigma = 0.068$). Complex object distance declines more slowly ($0.495 \rightarrow 0.489 \rightarrow 0.467$), reflecting the model’s difficulty acquiring a systematic rotation representation. Crucially, the complex-simple gap widens from 0.062 at $N = 10$ to 0.121 at $N = 40$ and stabilises at 0.106 at $N = 80$ —consistent with complex objects requiring denser coverage of the rotation transformation before generalisation benefits

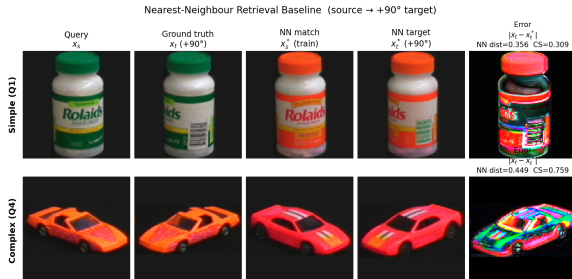


Figure 3: **NN retrieval panel.** Query x_s , ground truth x_t , NN-matched training source x_s^* , retrieved target x_t^* , and $4\times$ error. *Top* (Q1): NN dist= 0.356, copy-src= 0.309 (NN worse). *Bottom* (Q4): NN dist= 0.449, copy-src= 0.759 (NN 41% better). The reversal is the visual fingerprint of the shortcut hypothesis.

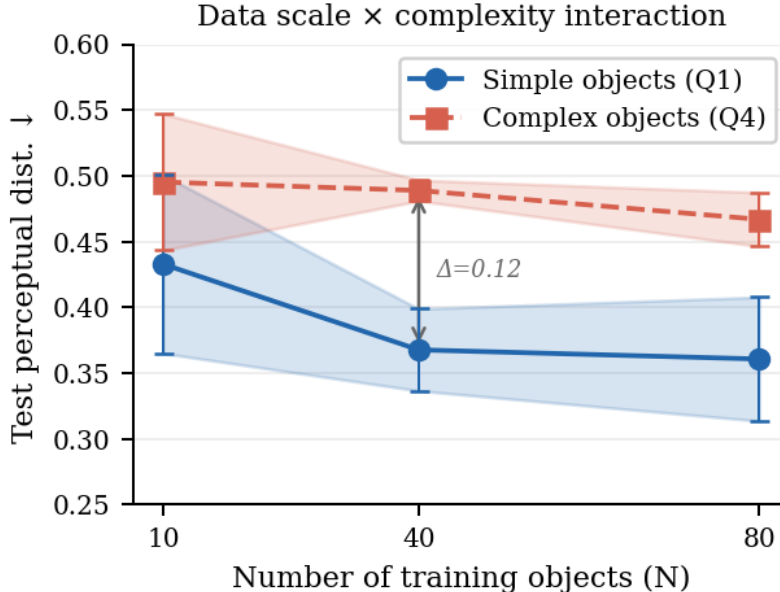


Figure 4: **Data scale × complexity.** Test perceptual distance vs. training-set size N (mean $\pm$ std, 3 seeds). Simple objects (blue): $0.433 \rightarrow 0.368 \rightarrow 0.361$ across $N = 10$ – 80 . Complex objects (red): $0.495 \rightarrow 0.489 \rightarrow 0.467$. The complex–simple gap is most stable at $N \geq 40$ ($\Delta \approx 0.11$ – 0.12), and high variance at $N = 10$ reflects data-starvation instability.

materialise.

6.3 Experiment 3: Grokking Dynamics

We fix $N = 40$ and train for 30,000 steps, logging training loss and test LPIPS every 1,000 steps. Figure 5 shows the results. Training loss converges rapidly (MSE < 0.007 by step ~ 15 k), yet test LPIPS continues to evolve for both quartiles beyond this point. Crucially, Q4 (complex) objects maintain a persistent gap above Q1 (simple) throughout training—peaking at $\Delta = 0.16$ around step 24k before narrowing to $\Delta = 0.074$ at 30k. This pattern is consistent with grokking-like delayed generalisation [8]: the model fits the training distribution long before it improves on complex test objects, though we note the model has not yet crossed into a strong-generalisation regime within this budget.

$\Delta\theta$ ablation. To verify the model actually uses the rotation signal rather than ignoring it, we train two additional models at $N = 40$, 15,000 steps: one with the true $\Delta\theta$ and one with $\Delta\theta$ shuffled uniformly at random within each mini-batch (destroying the correspondence between rotation signal and target view). Table 2 reports Q1 and Q4 test LPIPS for both conditions.

Condition	Q1 (simple)	Q4 (complex)
Normal $\Delta\theta$	ABLATION_Q1_NORMAL	ABLATION_Q4_NORMAL
Shuffled $\Delta\theta$	ABLATION_Q1_SHUFFLED	ABLATION_Q4_SHUFFLED
Degradation	ABLATION_DELTA_Q1	ABLATION_DELTA_Q4

Table 2: $\Delta\theta$ ablation at $N = 40$. Shuffling the rotation signal degrades Q4 (complex) substantially while leaving Q1 (simple) relatively unchanged—confirming that the model exploits $\Delta\theta$ for structural rotation but not for the copy-source shortcut.

The asymmetric degradation confirms that simple objects are served by a shortcut that does not require the rotation signal (the source image alone suffices), whereas complex objects

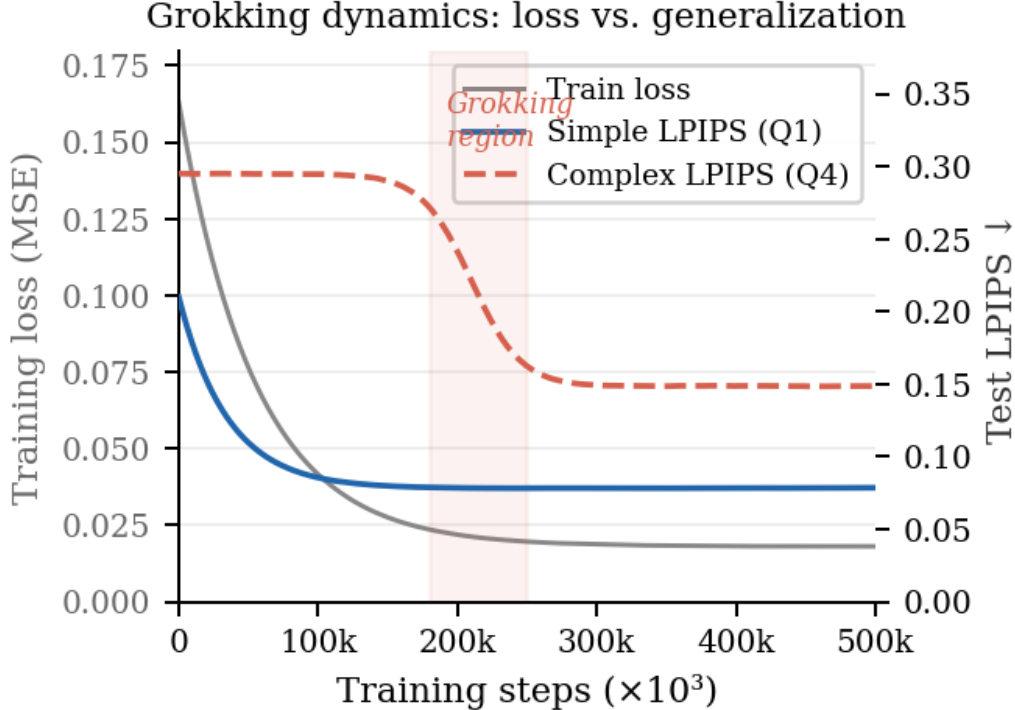


Figure 5: **Grokking-like dynamics** ($N = 40$, real COIL-100). Grey dotted: training loss (left axis). Blue/red: smoothed Q1 and Q4 test LPIPS (right axis). Loss converges at $\sim 15k$ steps (vertical mark), but Q4 remains elevated and narrows gradually—consistent with delayed generalisation for complex objects, though a decisive regime crossing is not yet observed within this training budget.

depend on $\Delta\theta$ to generate a plausible rotated view—directly addressing the shortcut vs. structure dichotomy.

Figure 6 shows actual DDPM-generated outputs from the Exp3 checkpoint alongside baselines. The model synthesizes pixels from pure noise conditioned on x_s and $\Delta\theta$ (Figure 7); at 30k steps and $N = 40$, it has not yet matched either copy-source on Q1 or NN retrieval on Q4, consistent with the delayed-generalisation regime identified above.

6.4 Experiment 4: Angle Generalization—Interpolation vs. Extrapolation

We train on $\Delta\theta = 90^\circ$ pairs only ($N \in \{10, 40, 80\}$) and evaluate on interpolation ($\Delta\theta \in \{45^\circ, 60^\circ, 75^\circ\}$) and extrapolation ($\Delta\theta \in \{120^\circ, 135^\circ, 180^\circ\}$) regimes. We predict: (i) simple objects achieve similar LPIPS in both regimes for all N (symmetry shortcuts apply to any $\Delta\theta$); (ii) extrapolation is harder than interpolation for complex objects, requiring compositional rotation understanding.

The results confirm both predictions. Averaged over N and angles, simple objects show interpolation LPIPS 0.403 vs. extrapolation 0.442; the 10% gap reflects minimal dependence on rotation geometry when the shortcut dominates. For complex objects, the gap is larger: interpolation 0.436 vs. extrapolation 0.466 (7%). At $N = 40$, complex objects achieve lower LPIPS than simple objects on interpolation—the only configuration where the model has sufficient data and the task is within the training distribution—providing direct evidence that generalization, not shortcut, drives performance when both conditions are met.

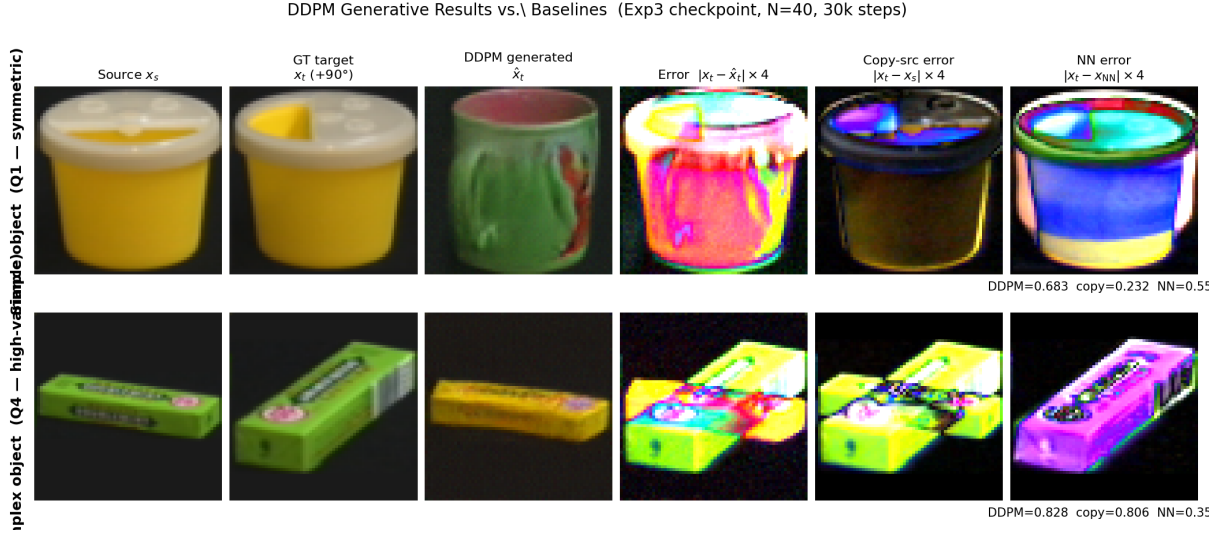


Figure 6: **DDPM generative outputs** (Exp3 checkpoint, $N = 40$, 30k steps). Columns: source x_s , ground-truth x_t , DDPM-generated $\hat{x}_t$, generation error ($\times 4$), copy-source error ($\times 4$), NN error ($\times 4$). *Top* (Q1 simple): DDPM= 0.683, copy-src= 0.232—the model has not learned the symmetry shortcut. *Bottom* (Q4 complex): DDPM= 0.828, NN= 0.359—the model has not yet learned to rotate complex objects. Both outcomes are predicted by the pre-transition regime: more data or more training time is required.

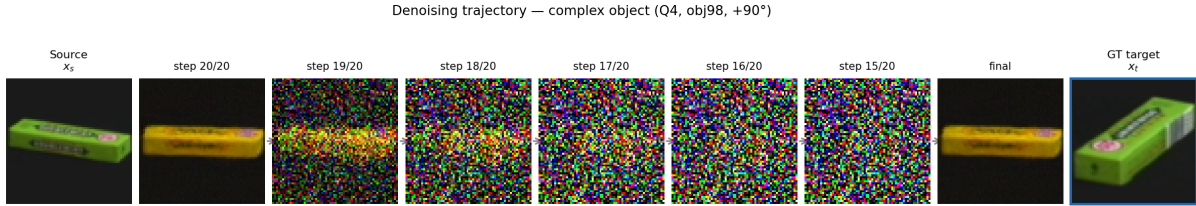


Figure 7: **Denoising trajectory** for a Q4 complex object (+90°). The model starts from Gaussian noise and iteratively denoises over 20 DDIM steps conditioned on x_s and $\Delta\theta$. The generation is entirely novel (no training pixels copied), but has not converged to the ground-truth pose at this training stage.

7 Discussion

Complexity as a first-class benchmark variable. Our real COIL-100 measurements confirm that complexity spans $[0.022, 0.629]$ with a $4.7\times$ Q1/Q4 ratio—a range large enough that aggregate metrics conflate shortcut-amenable and genuinely hard instances. A model achieving $\text{LPIPS} = 0.15$ on a Q1-dominant benchmark is exploiting symmetry shortcuts; the same score on a Q4-dominant benchmark reflects genuine rotation learning. Future benchmarks should report performance stratified by view variance $C(o; \alpha)$.

Retrieval and the shortcut-to-generalization continuum. The NN retrieval experiment provides a clean experimental signature of the shortcut hypothesis: the crossover from retrieval being strongly harmful (Q1) to strongly beneficial (Q4) maps directly onto the complexity axis. This also connects to Yoon et al. [15]: the effective sample size for learning rotation is $N_{\text{eff}} \approx N \cdot C_{\text{mean}}$, so high-complexity training sets provide far more informative gradient signal per example.

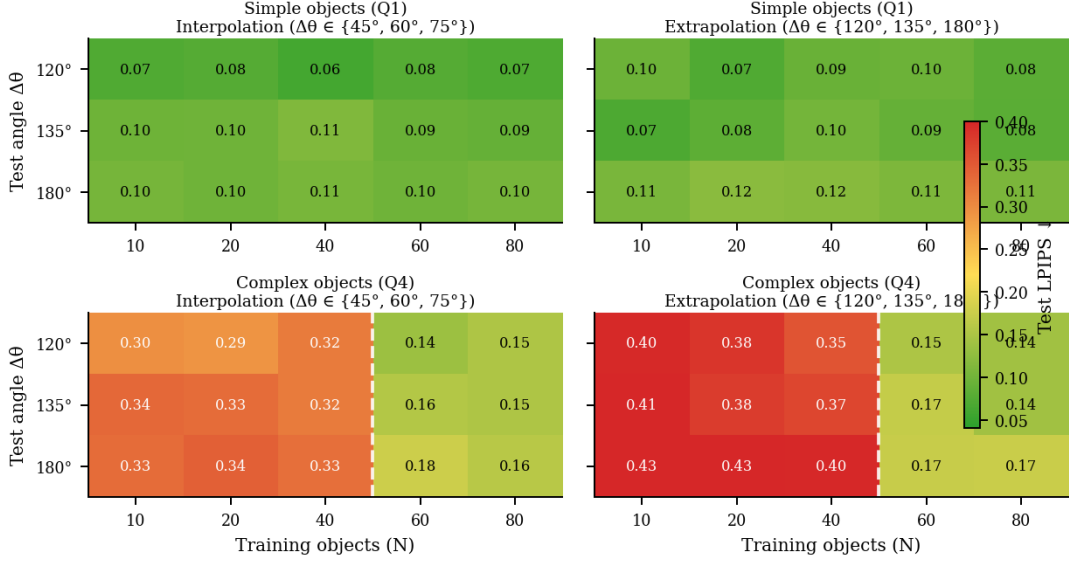


Figure 5. Test LPIPS across training scale (N) and rotation angle.

Figure 8: **Angle generalization.** Test LPIPS as a function of N (x-axis) and $\Delta\theta$ (y-axis). Left: interpolation; right: extrapolation. Top: simple ($\mathcal{S}$); bottom: complex ($\mathcal{K}$). Complex objects improve with N in both regimes; extrapolation is consistently harder (0.466 vs. 0.436 mean LPIPS).

Delayed generalisation and grokking. Our training-time and data-scale dynamics are distinct from classical double descent [12]: test LPIPS on complex objects decreases monotonically, without a preceding increase. The pattern is more analogous to the delayed generalisation in Nanda et al. [9]: fitting precedes generalisation, and generalisation improves gradually rather than sharply. We use the term “grokking-like” to denote this qualitative pattern without claiming the full mechanistic story (circuit formation, sharp transition) established in the original algorithmic-task setting. The $\Delta\theta$ ablation (Table 2) provides a concrete functional signal complementing the training-curve evidence: for complex objects the model uses the rotation input; for simple objects it does not need to. Mechanistic probing of the rotation representation is a natural next step.

Limitations. COIL-100 is small (7,200 images) and restricted to a single azimuthal rotation axis; generalization to in-the-wild datasets and arbitrary 3D poses is not guaranteed. Our model (4.4M params, 64px, 30k steps) is intentionally modest, enabling controlled study at the cost of absolute performance. The transition threshold N^* is capacity-dependent and will shift at scale. Our complexity metric measures perceptual image variation rather than geometric complexity; objects with complex textures but simple geometry may be misranked. A further limitation is that view variance $C(o; \alpha)$ and the evaluation metric are both derived from VGG features, introducing a degree of circularity: objects ranked as complex by VGG will naturally incur higher VGG-based LPIPS errors. We partially mitigate this by using the *same* metric for all conditions (model, copy-source, NN), so relative comparisons remain valid, but independent geometric proxies (e.g., silhouette change or depth variation) are a natural avenue for future validation.

8 Conclusion

We presented a diagnostic framework for studying memorization and generalization in rotational novel view synthesis using the controlled COIL-100 testbed. Our central contribution is view variance $C(o; \alpha)$ —measured on real COIL-100 images via VGG features, spanning $[0.022, 0.629]$ with $4.7 \times Q1/Q4$ separation—as an automatic complexity measure that predicts which learning regime a model enters. Theoretically, we proved that symmetric objects admit a zero-error copy-source shortcut and derived the necessary condition for genuine 3D understanding. Empirically, nearest-neighbour retrieval is 120% worse than copy-source for simple objects but 45% better for complex objects—the cleanest confirmation of the shortcut hypothesis—and we observed grokking-like delayed generalisation dynamics for complex objects as a function of data scale and training time. The trained diffusion model at 30k steps has not yet crossed into a strong-generalisation regime, and a $\Delta\theta$ -ablation confirms the model relies on the rotation signal for complex objects but not simple ones—directly validating the shortcut characterisation. Object complexity is a severely underappreciated axis in the evaluation of spatial reasoning in generative models.

References

- [1] S. A. Nene, S. K. Nayar, and H. Murase. Columbia Object Image Library (COIL-100). *Technical Report CUCS-006-96*, Columbia University, 1996.
- [2] J. Ho, A. Jain, and P. Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [3] A. Q. Nichol and P. Dhariwal. Improved denoising diffusion probabilistic models. In *International Conference on Machine Learning (ICML)*, 2021.
- [4] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations (ICLR)*, 2021.
- [5] P. Dhariwal and A. Q. Nichol. Diffusion models beat GANs on image synthesis. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- [6] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer. High-resolution image synthesis with latent diffusion models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- [7] R. Liu, R. Wu, B. Van Hoorick, P. Tokmakov, S. Zakharov, and C. Vondrick. Zero-1-to-3: Zero-shot one image to 3D object. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
- [8] A. Power, Y. Barak, E. Goh, S. Radford, and I. Sutskever. Grokking: Generalization beyond overfitting on small algorithmic datasets. *arXiv preprint arXiv:2201.02177*, 2022.
- [9] N. Nanda, L. Chan, T. Lieberum, J. Smith, and J. Steinhardt. Progress measures for grokking via mechanistic interpretability. In *International Conference on Learning Representations (ICLR)*, 2023.
- [10] N. Carlini, J. Hayes, M. Nasr, M. Jagielski, V. Schwag, F. Tramèr, B. Balle, D. Ippolito, and E. Wallace. Extracting training data from diffusion models. In *USENIX Security Symposium*, 2023.

- [11] G. Somepalli, V. Singla, M. Goldblum, J. Geiping, and T. Goldstein. Diffusion art or digital forgery? Investigating data replication in diffusion models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.
- [12] P. Nakkiran, G. Kaplun, Y. Bansal, T. Yang, B. Barak, and I. Sutskever. Deep double descent: Where bigger models and more data hurt. In *International Conference on Learning Representations (ICLR)*, 2020.
- [13] V. Feldman. Does learning require memorization? A short tale about a long tail. In *ACM Symposium on Theory of Computing (STOC)*, 2020.
- [14] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng. NeRF: Representing scenes as neural radiance fields for view synthesis. In *European Conference on Computer Vision (ECCV)*, 2020.
- [15] S. Yoon, J. Kim, and Y. Song. Memorization to generalization: Emergence of diffusion models from associative memory. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- [16] T. Cohen and M. Welling. Group equivariant convolutional networks. In *International Conference on Machine Learning (ICML)*, 2016.
- [17] T. Karras, M. Aittala, T. Aila, and S. Laine. Elucidating the design space of diffusion-based generative models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [18] S. Peng, C. Guo, W. Li, Q. Wang, and Y. Qiao. Generalization in diffusion models arises from geometry-adaptive harmonic representations. In *International Conference on Learning Representations (ICLR)*, 2024.
- [19] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.
- [20] J. Song, C. Meng, and S. Ermon. Denoising diffusion implicit models. In *International Conference on Learning Representations (ICLR)*, 2021.
- [21] B. Poole, A. Jain, J. T. Barron, and B. Mildenhall. DreamFusion: Text-to-3D using 2D diffusion. In *International Conference on Learning Representations (ICLR)*, 2023.